Chapter 4
Computer Memory

Memory is used to hold binary strings of data that is going to be manipulated by CPU. There are two major classes of memory: non-volatile and volatile. Non-volatile memory keeps the content even if the computer is off or power is gone. The set up data held in CMOS uses this technique. In volatile memory, the data is lost when the computer loses power. It keeps data as long as only computer is on. Random Access Memory is good example of volatile memory. By default, when we say memory, we mean RAM.

RAM
Whenever you run a program on your computer, that program first loads into RAM before anything happens. It holds temporary instructions and data for manipulation when the system is running.

It is called Random Access because CPU can access or place data to and from any part of RAM on the system. RAM come in wide variety of modes defined by physical size, access speed, electrical set up, and width of the bus measured in bits. The bit width of RAM defines the amount of information your CPU can access or write to your RAM within one cycle.

Types of RAM
There are two types of RAM based on the way they keep data:

- ✓ Static RAM (SRAM)
- ✓ Dynamic RAM (DRAM)

1) DRAM
This memory type is dynamic. Because of this, it must be constantly refreshed periodically(every few milliseconds). Otherwise the memory will drain and the data is lost. During the process of refreshing, the CPU can't access the memory. It is called wait state. Refreshing caused DRAM to be slower than SRAM. They also use much power than SRAM. But because of their cheapness, they are the primary RAM in all computers.

The memory cells in DRAM use tiny capacitors that retain charge. Capacitors are devices that can keep charge for some time until discharged. They use one transistor per bit. Because of this they can be densely packed. This allows more memory capacity per chip than other types of memory.

Note: there are 256million transistors in 256MB RAM chip. Compare this with Pentium II which has 7.5 million transistors.

Disadvantage:
DRAM is slow, much slower than CPU. This affects the speed of CPU. It may take as much as 10% of CPU time.

2) SRAM

It is called because it does not require periodic refreshing unlike DRAM. It is much faster than DRAM and able to keep pace with processor. As long as there is power, SRAM keeps what is stored.

SRAM uses transistors for storage purpose, no capacitors are used. Because of this, no recharging required. It uses 6 transistors (flip flops) per bit. DRAM is lower in density which means they are larger physically and store fewer bits overall. The high number of transistors used per bit makes it more expensive and physically larger. This prevented its wide use as computer memory. SRAM is used to build cache memory.

Type	speed	Density	cost
DRAM	slow	High	low
SRAM	fast	Low	high

RAM Modules

There are two types of RAM modules (memory packages)

- ✓ Single Inline Memory Module(SIMM)
- ✓ Dual Inline Memory Module(DIMM)

1) SIMM

SIMMs are available in two forms:

- ✓ 30 pin and
- ✓ 72 pin

The 30 pin SIMMs are available in sizes 1-16MB. They transmit one byte of data (8 bits) at a time. They are older standard.

The 72 pin SIMMs are available in 1-32MB. They transmit 4 bytes (32 bits) at a time.

SIMMs are inserted into a socket on the motherboard that will tightly hold them. They are available in single sided or double sided which means DRAM chip is available in both sides of the SIMM or not.

30 pin—single side

72 pin—both single and double sided

2) DIMM

DIMM is used in latter versions of computer system. DIMM has 168 pins. They transmit 64 bits of data at a time. They are becoming very popular. They are not available in smaller sizes such as 1MB or 4MB. They are available in capacities from 8 MB to 256MB.

DIMMs are inserted into special sockets on the motherboard similar to those used for SIMMs. They differ from SIMM in:

- ✓ they are larger and wider
- ✓ SIMMs have contact on both sides of the circuit board but they are tied together. A 30 pin SIMM has 30 contacts on each side of the circuit board but each pair is connected. This is true for 72 pin SIMMs. DIMMs, however, have

different connections on each side of the circuit board. So, 168 pin DIMM have 84 contacts on each side and they are not redundant.

A small version of DIMM is seen which is called Small Outline DIMM (SODIMM). They are used for laptop PCS. Two types:
144 pin SODIMM-64 bits wide
72 pin SODIMM-32 bits wide

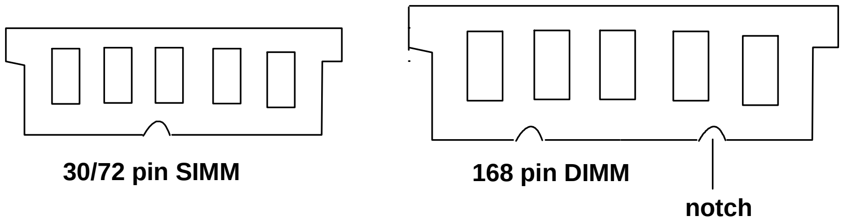

Fig SIMM and DIMM comparison

Asynchronous and Synchronous DRAM
Conventional DRAM, the type that has been used in PCs since the original IBM PC days, is asynchronous. In asynchronous memory, the memory is not synchronized to the system clock. A memory access is begun and at certain period of time later, the memory value appears on the bus. It works in low speed memory bus system but not suitable for use in high speed systems.

A newer type of DRAM called synchronous DRAM (SDRAM) is synchronized to the system clock. As a result, timing is much tighter and better controlled. This memory is much faster than asynchronous DRAM. It is suitable for high speed systems.

Memory Banks
Manufacturers arrange memory slots on motherboard in electronic groups of one, two, or four RAM slots according to the type of RAM and processor. This grouping is called memory banks. The memory bank matches the processors' data bus.

the number of slots to form a bank= $\frac{\text{bus width of CPU}}{\text{bus width of RAM slot}}$

E.g. CPU bus=64 bits
RAM bus=8 bits
Bank=64/8=8 slots. 8 slots form one bank in such case.

The basic rule of bank is you need to fill a bank completely with identical RAM modules. Banks should be filled completely or left completely empty. You should fill at least one bank on the computer for it to work properly. For 64 bit CPU, one 168 pin DIMM slot makes a bank. For the same CPU, two 72 pin SIMM slots form a bank.

Types of RAM

There are many types of RAM:

1) Conventional DRAM:

the oldest and slowest DRAM technology is called regular or conventional DRAM. It uses standard memory addressing method, where first the row address is sent to memory and then column address is sent.

2) Fast Page Mode RAM (FPM RAM)

FPM is slightly faster than conventional DRAM. This memory works by sending row addresses just once for many accesses to memory locations in near each other. This improves access time. In older ones, you have to specify the row and column address. Despite its name, FPM is the slowest memory technology used in modern PCs.

3) Extended Data Out RAM (EDORAM)

it is the most common type of asynchronous DRAM. It is slightly faster than FPM. This is because its timing circuits are modified. The architecture allows to simultaneously read new data while discharging the old. It allows to begin a new column address instruction while it is reading data at the current address. To use EDO, your motherboard chipset must support it. It has 72 pins (SIMM module). It has one notch in the center.

Fig EDO RAM

4) Synchronous DRAM (SDRAM):

This is a relatively new and different kind of RAM. It differs from earlier types in that it is tied to the system clock i.e. synchronous to system clock. It is a great improvement over FPM or EDO RAM. It delivers data in high speed bursts. It also runs at a speed of system bus due to synchronization. It is new standard RAM for modern PC.

It is sold in DIMM form and is often rated by megahertz speed rather than nanosecond cycling of previous RAM types. It has 168 pins and two notches.

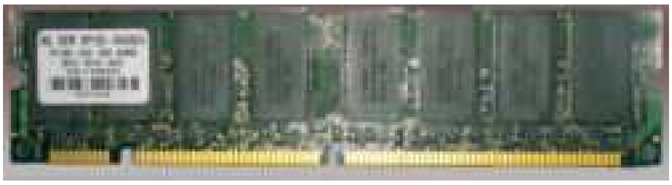

Fig SDRAM

5) Double Data Rate SDRAM(DDR SDRAM)
DDR works the same way as SDRAM. It doubles bandwidth of memory by transferring data twice per cycle—on both the rising and falling edges of the clock signal. Clock changes from 0 to 1 and from 1 to 0. Both changes trigger data transfer in DDR RAM which is not the case in previous RAM types. Older system use only one edge.

DDR SDRAM doubles data transfer rate. It has one notch in the center

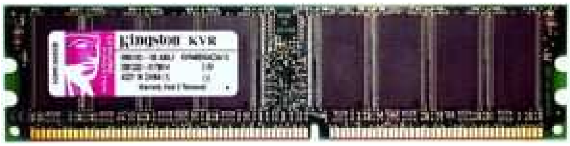

Fig DDR RAM

6) Rambus DRAM (RDRAM)
This is a radical new memory design found in high end PC systems. It was first developed for game technology and it became popular.

RDRAM transfers only 16 bits at a time. But they transfer at a much faster speed. They are called narrow channel device unlike wide channel systems which transmit as much as processor's data bus i.e. they are not as wide as processor's data bus. They run at 800MHZ. The overall throughput is 800x2=1.6GB per second. This is twice as fast as that of SDRAM. It has two notches in the center and 168 pins.

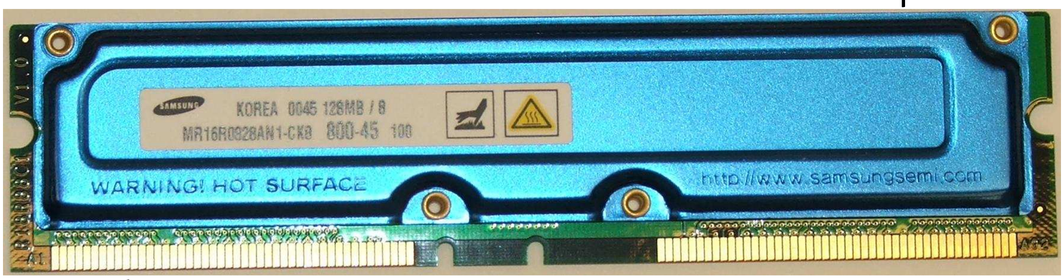

Fig Rambus RAM

The narrowing of the channel allows the running at high speeds. It transfers data on both rising and falling edges of clock like DDR.

7) Video RAM (VRAM)
Modern video adapters use their own specialized RAM that is separate from main system memory. The demands placed on video memory is far greater than placed on RAM. They have to be accessed at faster speeds. They allow the memory to be accessed by CPU and video card's refreshing circuitry at the same time. This is called dual porting. It is used on graphics adapters.

8) Windows Accelerator Card RAM (WRAM)

It is designed for two reasons:

- ✓ To speed up video(faster video)
- ✓ To speed up Windows operating system

It could be read and written at the same time. As video technology became more advanced, VRAM could not provide high quality video. WRAM increased speed and provided excellent video. Most video cards now use SDRAM for video RAM.

Cache Memory

Cache is a high speed buffer made up of SRAM that directly feeds the processor. It runs at the speed close or equal to the processor. This enables the CPU to use its full potential by getting data and instructions from the cache memory without any wait state. RAM is slow and slows down the processor because the processor has to wait during memory refreshing. To avoid wait states, data is preloaded from RAM (DRAM) to cache memory and the CPU can continue to process during wait states by getting data from cache memory though RAM is not available.

There are two types of cache memory:

- ✓ L1 cache
- ✓ L2 cache

L1 Cache (Level 1)

It is directly built into the processor and is actually part of the processor. Starting from 486, there are internal caches in CPU. All commands for the processor go through this cache. It stores a backlog of commands so that if wait state is encountered, the CPU continues to process using commands from the cache.

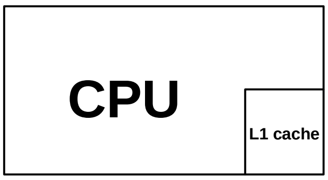

Fig CPU and L1 cache

L2 Cache

This cache is mounted directly on the motherboard, outside CPU. It is called Level 2 cache. It is same as L1 cache but larger. It can be expanded (add additional capacity) on some motherboards. If you install additional L2 cache, check the CMOS set up and enable the cache. Some L2 caches are found on the processor itself. L2 caches located on the motherboard run at the speed of motherboard.

Memory Usage

The system memory though it is often referred to as a single number, in fact is broken into several different areas. This is a design in legacy systems built into earlier versions of IBM PC, and the versions of DOS that run on them.

The original Intel processor could use only 1MB of RAM to the maximum. The original IBM PC allowed only 640KB of 1MB RAM for direct use. MS-DOS applications were written to conform to this. The IBM pc divides the memory into two areas:

- ✓ Conventional memory
- ✓ Reserved memory

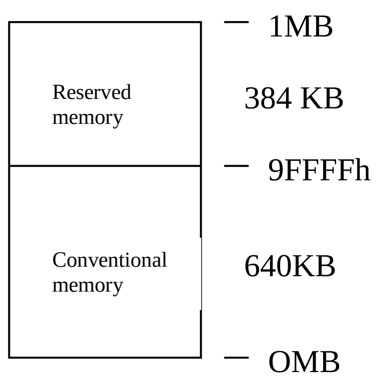

Fig memory usage

Conventional Memory

- ✓ Takes the first 640KB of the memory
- ✓ It is used for operating system and application programs
- ✓ DOS program use only this memory

Reserved Memory

The remaining area out of 1MB not occupied by conventional memory is called reserved memory. It is 384KB in size. It is used for BIOS, Video RAM, ROM, and so on.

Memory Division

As a result of a decision made in earlier PCs, memory is broken into four basic pieces(with some pieces being divided further):

- ✓ Conventional memory
- ✓ Upper memory area(UMA)
- ✓ High memory area(HMA)
- ✓ Extended memory

Conventional Memory

The first 640KB of system memory is called conventional memory. This is a place where DOS and DOS programs conventionally run. Originally, this is the place where programs run. Without special software support, DOS can't run programs that are not in this special area. Conventional memory occupies the address 00000h to 9FFFFh.

The upper memory area that are not used by ROM or for Video RAM are available for use by other programs. But they are very small and can't be used by programs. They are used for loading memory resident programs and drivers. To use this area, driver programs are required to access them. The driver used in DOS environment is EMM386.exe. HIMEM.SYS is also used.

Upper Memory Area (UMA)

The upper 384KB of the first Megabyte of RAM directly above conventional memory is the UMA. This area is reserved for video RAM, BIOS, motherboard BIOs, etc. It starts from the address A0000 through FFFFF.

It is used as follows:

- ✓ First 128 KB is called video RAM. It is reserved for use by video adapters. Address range A0000-BFFFF. Anything displayed on the screen is stored here.
- ✓ The next 128KB is reserved for the adapter BIOS that resides in ROM chips on some adapter boards plugged into bus slots. E.g network adapters (NIC), VGA-compatible video adapters, etc. The address range is C0000-DFFFF.
- ✓ The last 128KB is reserved for motherboard BIOS i.e. instructions stored in ROM chips. The bootstrap loader also resides here. The address range is E0000-FFFFF.

ROM Shadowing

One problem with ROM those used for system BIOS and Video BIOS is that it is slow. Access to BIOS codes are very slow relative to system memory (RAM). For this reason, system mirror the ROM code into RAM to improve performance. This is called ROM shadowing. ROM is copied to RAM and used from RAM. The RAM is assigned address originally used by ROM i.e. there is memory address overlap between RAM and ROM. This disables the ROM.

Extended Memory

In modern systems, the memory above 1MB is used as extended memory. The name comes from the fact that this memory was added as an extension to 1MB base that was the upper limit possible in previous systems.

Extended memory is not accessible under real mode. It is addressed only under protected mode, the native mode of modern processors starting from 286.Two ways to use extended memory:

- ✓ A full protected mode operating system like Windows NT can access extended memory directly
- ✓ Operating system or application that run in real mode(DOS, Windows 3.x, Windows 95) use an extended memory manager to access the extended memory. The most commonly used manager is HIMEM.SYS. This memory manager sets up extended memory based on e**x**tended **m**emory **s**pecification (XMS). XMS is a standard that PC programs use to access extended memory.

High Memory Area

High memory area is the first 64KB of extended memory. Technically, it is part of extended memory. It is the only part of extended memory that PCs operating in real mode could access/use. A software called A20 must be run to access this area. It is also possible to use HIMEM.SYS.

Troubleshooting Memory

If there is a memory problem in your computer, your computer does not display anything on the monitor the same as when CPU failed. When it is turned on, the BIOs gives you a beep sound-continous beep sound.

Upgrading and Replacing RAM

Before you buy a new chip for upgrading/replacing, check what type of RAM fits into the RAM slots of the motherboard. What type of RAM does the motherboard support?

- ✓ EDO RAM
- ✓ SDRAM
- ✓ DDR RAM
- ✓ RDRAM, etc

Buy one that is supported by the motherboard.

Steps to install RAM:

1. open the system following safety cautions.
2. check in which direction the memory fits into the slot i.e. it does not fit in the wrong direction. To determine direction, check the notches of the RAM and slot.
3. Properly align the RAM on the slot. On some systems especially older ones, the RAM is inserted in some angle to the motherboard, not straight. Newer RAMs fit perpendicular to the motherboard.
4. Push the RAM proportionally form both edges. A holder on both edges of the slot should hold the RAM.
5. Check if the RAM is fully inserted and does not move.